

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
2.671 Measurement and Instrumentation

Uncertainty Analysis Part 3: Comparing Populations

This is the third document created to teach you how to analyze and present experimental uncertainty. You should have already read Part 1, which introduced you to basic concepts in uncertainty analysis and showed you how to apply them to estimate uncertainty on parameters derived from measurements (such as average value and fit parameters for arbitrary functions fit to your data, including linear, polynomial, exponential, power law, etc.). Part 2 provides further information on curve fitting and model selection. **This document describes how to compare different samples and evaluate whether they are statistically the same or different, and if different, how to quantify the difference.** You will need to use the material in this document for the Soda Can Postlab Part 3. All three Uncertainty Analysis documents will be indispensable to you in the analysis of your Go Forth and Measure results, and the concepts will be further reinforced in future labs. This document can be found online in the Uncertainty page of the Wiki as a PDF file. We will give you pointers on how to perform calculations with three programs accessible to you: Logger Pro, Excel, and MATLAB. We have also created a “Slide Deck” version that extracts the essential information from this document.

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1. Discussion of Confidence Intervals and Probability (repeated from Part 1)

The following section was included in Uncertainty Analysis Part 1, but bears repeating here because it is relevant to this document as well. Confidence intervals were first described in the 1930s by Neyman [1]. However, there is still ongoing controversy regarding how to describe and interpret them. [2]

The most important thing to keep in mind is that reporting something “with 95% confidence” does **not** mean that there is a 95% probability that true mean is within your estimated confidence interval. The true population mean is (usually) not known, but if it is known, it is either contained within the confidence interval or it is not.

The correct statement is that if you performed the experiment with the same experimental procedure 100 times, and constructed confidence intervals from your sample measurements using the same procedure each time, you expect that 95 of the confidence intervals you construct would contain the true population mean. However, if you only perform the experiment (sample the population) one time, there is a chance that your results are one of the 5/100 cases where the confidence interval does **not** contain the true mean, in which case, there is a 0% probability that the mean is contained within your confidence interval.

This distinction can be very important if you are trying to estimate the probability that disastrous outcomes (e.g. a bridge collapsing, a medical treatment causing death) will occur given the results of your measurements and your estimated confidence intervals from a small set of experiments. Matters of safety typically require larger margins of safety, so 95% confidence intervals (representing 1/20 chance of random replication) may need to be elevated to 99.9999% confidence intervals (representing 1/1,000,000 chance of random replication) or higher.

If you are performing multiple independent tests, you may need to select a higher confidence level to avoid false positives. [3]

The best approach for reporting uncertainty is to simply give your mean with the confidence interval, and not make claims about the “true” value of the mean. For example, say “The measured pressure for Coke was 362 ± 13 kPa” rather than using the common misconception that “The level C of a confidence interval gives the probability that the interval produced by the method employed includes the true value of the parameter”, a misconception that was repeated in online material from a Yale statistics class in the late 1990s. [4]

2. Comparison of Results from Different Measurements (Student’s t-test)

A common question, particularly for Go Forth and Measure projects, is whether there is a statistically significant difference between two measurement sets, or whether a particular set of measurements has a mean that is statistically different from (or the same as) a given value. This question can be answered using the “Student’s t-test Method”. This method was developed by W.S. Gossett at the Guinness Brewery in 1908: Student is simply a pseudonym used since the brewery viewed their use of statistics to evaluate the quality of their beer as a trade secret. [5]

The following discussion assumes a Gaussian probability distribution as shown in Fig. 1

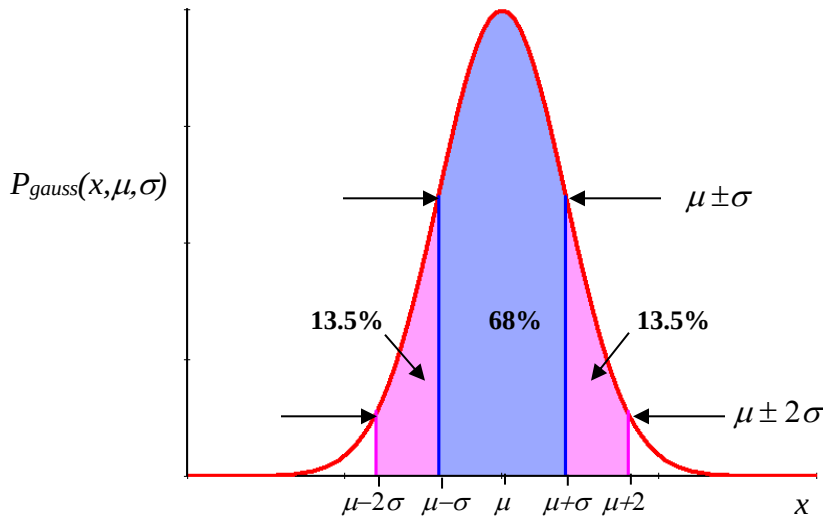


Figure 1: Gaussian probability distribution with mean μ and standard deviation σ . The blue shaded area indicates that 68% of the values fall within $\pm 1\sigma$ of the mean, and the combination of the blue and pink shaded areas indicates that 95% of the values fall within $\pm 2\sigma$ of the mean. (Fig. 4 of Uncertainty Analysis Part 1)

A proper discussion of significance testing requires the introduction of some new terms, which we will do by first introducing the **two-sample t-test**, and then follow up by extending the method to a **one-sample t-test**.

Imagine two populations of values, Population 1 and Population 2, that have unknown mean values μ_1 and μ_2 and unknown variances. To compare the two populations, you could observe a sample of the values from each, Sample 1 from Population 1, and Sample 2 from Population 2. The **two sample t-test** compares these two samples of values to determine the probability that they come from underlying populations with the same or different mean values.

The **two sample t-test** is defined in terms of two hypotheses, the **null hypothesis** and one of three types of alternate hypothesis, which you would select based on your situation.

The **null hypothesis** is: $\mu_1 = \mu_2$

The **two-tailed** alternate hypothesis is: $\mu_1 \neq \mu_2$

The upper **one-tailed** alternate hypothesis is: $\mu_1 > \mu_2$

The lower **one-tailed** alternate hypothesis is: $\mu_1 < \mu_2$

The “tails” in the descriptions of the alternate hypotheses refer to the Gaussian distribution shown in Fig. 1. As an example, the “upper tail” that includes 2.5% of the population is given by the region with $x > \mu + 2\sigma$, and the corresponding “lower tail” is the region with $x < \mu - 2\sigma$, which also includes 2.5% of the population. The distinction between **two-tailed** and **one-tailed** t-tests is discussed in the next subsection.

To answer the research question and accept or reject these hypotheses, we rely on the **t-value**. The **t-value** is a measure of extremeness, which compares the difference between the means of the two samples, accounting for the variation within the samples. Equation 1 defines the t-value, t , where μ_{sample1} is the mean value of Sample 1; σ_1 is the standard deviation of Sample 1; and n_1 is

the number of points in Sample 1. Similarly, μ_{sample2} , σ_2 , and n_2 , are the mean, standard deviation, and number of points in Sample 2, respectively.

$$t = \frac{\mu_{\text{sample1}} - \mu_{\text{sample2}}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (1)$$

Large magnitude *t*-values have low probability. By inspection of Eq. (1), you can observe that large magnitude *t*-values happen when the sample means are significantly different, and/or when the sample variations are small, and/or when the number of points are large.

The function of the *t*-test is to allow you to determine the level of statistical significance of the observed difference between your samples. The result of the *t*-test is usually expressed as a ***p*-value**, also called the “significance level”, which represents the probability of obtaining results that are at least as extreme as the results actually observed, assuming that the null hypothesis is correct.

For example: suppose that the observed mean value μ_{sample1} is different from μ_{sample2} , and we want to know if that observed difference is statistically significant. If a ***two-tailed t-test*** yields a ***p*-value** of 0.2, that implies that there was a 20% chance that we could have measured these values of μ_{sample1} and μ_{sample2} even if the true means of the underlying populations were equal. That is not a very convincing argument that the means are in fact different, so we accept the null hypothesis and conclude that the means are statistically indistinguishable based on the evidence from the sample data sets.

If on the other hand, the two-tailed *t*-test yields a ***p*-value** of 0.00001, the chance that we could have measured this value of μ_{sample1} and μ_{sample2} if the true means of the underlying populations were actually equal is extremely small. We therefore reject the null hypothesis and can state with 99.999% confidence that the true means are not equal. We have used here the fact that the **confidence level $\equiv 1 - p$. In 2.671, the standard level of confidence is 95%, which corresponds to a *p*-value of 0.05.**

A ***one-sample t-test*** is used if you have only a single data set (i.e. a single set of sample measurements), and would like to compare the population underlying this sample to a fixed value.

This type of test may be useful if you are comparing a set of measurements to a target value such as a theoretically modelled value, or a specified limit such as a legal limit (e.g. maximum allowable emission limit), a practical limit (e.g. the current limit of an electrical fuse) or an acceptance criterion in a quality system (e.g. all foot-long subs must be at least 0.3048 m in length).

2.1 Two-Tailed or One-Tailed T-Test?

The ***two-tailed t-test*** asks the question whether the means are different, not which one is greater than the other. A ***p*-value = 0.05** would result if the measured value fell in either the upper or lower tail, outside the pink regions shown in Fig. 1. This result indicates that with 95% confidence, $\mu_1 \neq \mu_2$.

If the two-tailed *t*-test *p*-value is 0.05, and the measured result happens to fall in the upper tail, a ***one-tailed t-test*** testing the upper tail alternate hypothesis would result in a *p*-value of 0.025 (half the two-tailed value). You might think that this indicates that $\mu_1 > \mu_2$ with 97.5% confidence based on the measured values in Sample 1 and Sample 2, i.e. that you know with higher confidence that

$\mu_1 > \mu_2$ than you do that $\mu_1 \neq \mu_2$. This seemingly illogical conclusion results from the fact that the action of doing a one-tailed t-test has excluded the possibility that in fact the sample mean is in the lower tail of the distribution – a possibility that is not negligible. You may do this exclusion **if and only if** you are positive that the Population 1 mean μ_1 can only be greater than μ_2 . For example, if you have a priori knowledge, ideally supported by fundamental principles, e.g. that the heater must always be hotter than the heated object, or that the arrival of a pool ball in the pocket must occur after the cue ball was struck, you can be sure that $\mu_1 > \mu_2$ and therefore would be justified in using a one-tailed t-test.

In some cases, you may elect to use a 1-tail test even if you don't have physical justification, if you are willing to accept the additional potential error of a 1-tail hypothesis in exchange for a stronger statistical statement. The additional potential error comes from the possibility that although one of your samples has a greater mean than the other sample, there is a finite probability that the means of the underlying populations actually have a reverse relationship. It is always possible to sample two populations in such a way that you derive a misleading relationship from the samples. If your 2-tailed p-value is infinitesimally small, you will incur very little error by performing a 1-tailed, rather than 2-tailed test. However, if your 2-tailed p-value is close to the 2.671 critical value of 0.05, switching to a 1-tailed t-test would not be a conservative approach.

In summary, you should always perform a **two-tailed t-test** unless you have strong scientific justification for knowing that a one-tailed test is appropriate. Figure 2 outlines the general approach to using the Student's t-test for significance testing, assuming the critical p-value = 0.05.

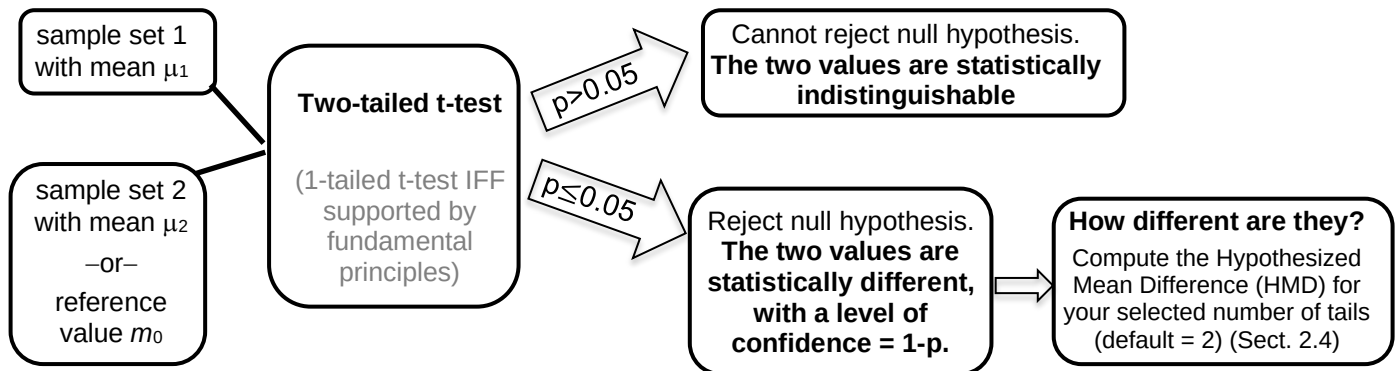


Figure 2: Flow diagram indicating how to perform a Student's t-test. The default is a 2-tailed t-test, unless you know a priori that one of the samples must have equal or greater mean than the other, presumably from fundamental principles. Sect. 2.2 describes how to perform the Student's t-test in Excel and MATLAB, and Sect. 2.4 describes how to compute the Hypothesized Mean Difference to answer the question of how different are the two results, in the case where the t-test indicates a statistically significant difference.

2.2 Computing the p-value

We will use a two-sample t-test as an example, as that is often the case in 2.671, when you want to compare the result of measurements made under two different conditions to evaluate the statistical significance of any observed differences. Suppose you have two sets of measurements, x_n and y_n , which have mean values $\bar{x}$ and $\bar{y}$, each of which may have different standard deviations and may include different numbers of data points. We can use the Student's t-test method to accept or reject the null hypothesis, by determining the p-value. Both Microsoft Excel and MATLAB can be used to compute the p-value. Please read the Excel section below even if you intend to use MATLAB, because it describes essential details of the calculation that apply to both environments.

Microsoft Excel can be used to compute the p-value – either through the function T.TEST (Mac and Windows) or preferably through the interactive t-test tool in the **Analysis Toolpak** (available on both Mac and Windows).

When you perform a t-test you need to decide how many “tails” to use and what “type” of t-test to perform. As described in Sect. 2.1, you should use a **two-tailed t-test** unless you have compelling reasons to know that the mean of one data set is by definition greater than the mean of the other data set, or unless the results of the 2-tailed test indicate extremely low probability that the underlying populations have the same mean (p-value $\ll 0.05$).

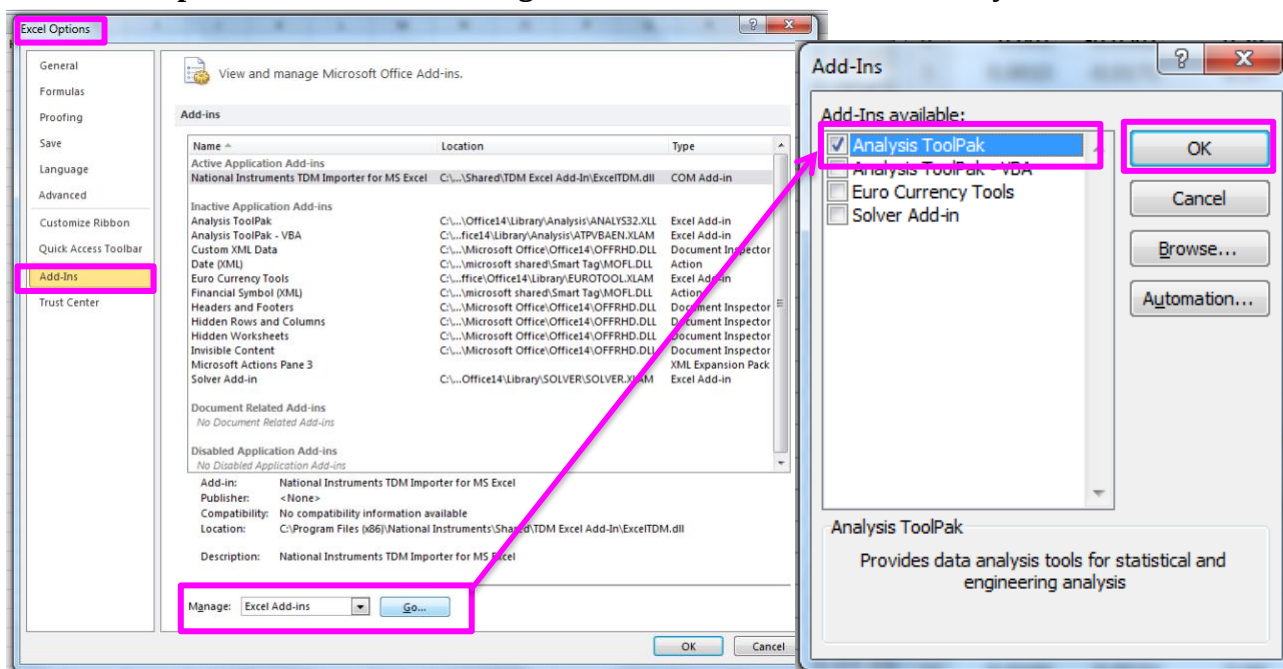
In most cases you will want to assume “**unpaired**” data sets; in other words, each value of x is not related to a particular value of y . This is the case when the two data sets are completely independent, as is true for the soda can example presented in Sect. 2.3. However, cases also arise when the data sets must be paired, i.e. if each x represents the state of a variable under one set of conditions (tumor size before treatment) and y represents the state of the same variable under another set of conditions (tumor size after treatment). In this case, each of the pair of samples could correspond to a different person, so each y value must be compared to its corresponding x value, and you should use a “paired” t-test. **Notice that a paired t-test can only be used if x and y have the same number of samples.** A case you will not often use is an unpaired t-test with equal variances – usually your two samples have different variances, as they arise from samples taken from different populations.

You will need to perform a Student’s t-test in the Lab 1 Postlab Part 3 in either Excel or MATLAB. Instructions for both codes are below.

Using the Student’s t-test in Excel

In order to perform the Student’s t-test in Excel, as well as perform other useful data analysis techniques such as making histograms, you must install the Data Analysis Toolpak in Excel (a free addition). To add it in Windows, do the following steps:

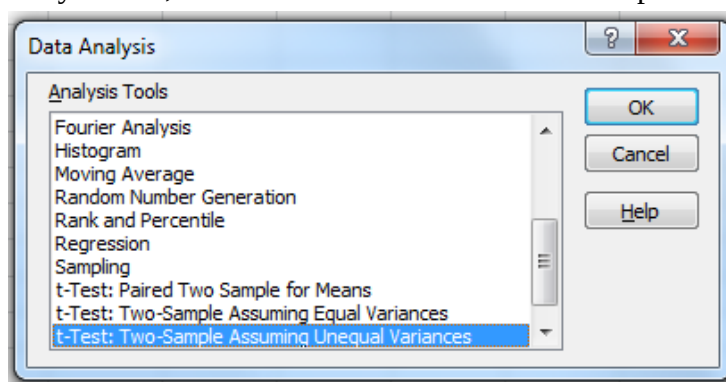
File – Options – Add-Ins – Manage: Excel Add-ins – Go, select Analysis ToolPak – OK



Go to the **Data** tab in Excel and you should now see **Data Analysis** on the far right of the tool bar

To use the Student's t-test in Excel, you need to have the two data sets in adjacent columns, not necessarily in the same columns as they are listed in the Soda Can Pressure Database. The instructions below are for performing the Lab 1 Postlab Part 3 analysis; simply modify as needed for other applications.

1. Create a new Tab in your Excel file and call it T-Test.
2. In the Data Analysis tool, scroll to the bottom and select the option shown below.



3. To make future steps easier, please select the data with the **higher average** for the **Variable 1 Range** and select the other data set as the **Variable 2 Range**.
4. Accept the default of zero for the **Hypothesized mean difference** and change the **Output options** to put the output in a blank part of the current tab, say in cell E1.

Assuming you are performing a two-tailed t-test (please refer to Sect 2.1), examine the value listed for $P(T \leq t)$ two-tail, henceforth referred to as the **p-value**. (Be careful that you don't read the p-value for the one-tail calculation, as that will be $\frac{1}{2}$ as large). If you find for your two data sets

that the *p-value* is **greater than 0.05**, it means that with 95% confidence, the two means are statistically indistinguishable; in other words, you **cannot** state with a confidence greater than 95% that the larger mean is equal to or greater than the smaller mean. The decision of what *p-value* is statistically significant is field dependent – in 2.671 we use 95% confidence ($p < 0.05$) as the limit of statistical significance. In other fields of study (e.g. medicine and drug development) the limit of statistical significance is often more stringent, requiring a *p-value* of 0.01 or less (at least 99% confidence). However, there is still merit in reporting the *p-value* to indicate your level of confidence in your results – for example, for a *p-value* of 0.10, you can state that the means are different with 90% confidence. It probably doesn't make sense to talk about confidence values less than 90%, but you may still report the *p-value* in your technical communication.

If on the other hand you find that the ***p-value* is less than 0.05**, you can state with a confidence of $> 95\%$ that the two means are different. **You would prefer to be able to state that one mean is greater than the other by some non-zero amount (with 95% confidence), so we will lead you through that calculation next. Make sure that you have correctly used the data set with higher mean as Variable 1, or this will not work.**

In order to find the hypothesized mean difference for which the *p-value* is less than or equal to 0.05, you use an iterative approach: simply guess and check different values in the hypothesized mean difference box in the T-Test popup in Excel. Start with a difference equal to the actual difference between the means – for this you should find a *p-value* greater than 0.05, i.e. a $< 95\%$ probability that the hypothesized mean difference equals the difference in mean values. Keep decreasing the value in this box until you reach a $P(T \leq t)$ two-tail value of approximately 0.05 or less. Please note that you should report the difference with the appropriate precision. In other words, if the mean of x is 375 ± 23 kPa, you should find the hypothesized mean difference with only 1 kPa numerical resolution, so you can correctly follow the Sig Fig rules of reporting a value to the same numerical precision as the uncertainty. If, of course, your uncertainty were 2.3 kPa, you would want to compute the hypothesized mean difference with 0.1 kPa resolution.

You will perform a t-test analysis in the 3rd Postlab for the Soda Can Experiment. If you are confused about how to apply the t-test, please ask 2.671 staff.

Using the Student's t-test in MATLAB

The function for two-sample t-tests in MATLAB is **ttest2**. The syntax

```
[h,p]= ttest2(x,y,Name,Value)
```

will return a value $h = 1$ if the test rejects the null hypothesis at the 95% confidence level, and 0 otherwise. The specific *p-value* as discussed above is also returned using this syntax. The **Name,Value** arguments are used to define the significance level ($\alpha = 0.05$ by default), define the number of tails (default = 2 tails, which in most cases is what you should use) and the variance type (usually unequal, since the two populations in most cases you study will not have equal variance). Refer to the excellent MATLAB Documentation for more help here: <https://www.mathworks.com/help/stats/ttest2.html> If you have not read the Excel discussion above, please do so if you are confused about how to correctly apply the MATLAB function.

We have provided building blocks of code on the MATLAB – Uncertainty Calculations page of the Wiki for performing t-tests and computing the Hypothesized Mean Difference (HMD).

2.3 Example: Coke vs Mountain Dew

As an example, we will look at the comparison of the historical data for Coke and Mountain Dew. The probability density functions for the two populations are given in Fig. 3.

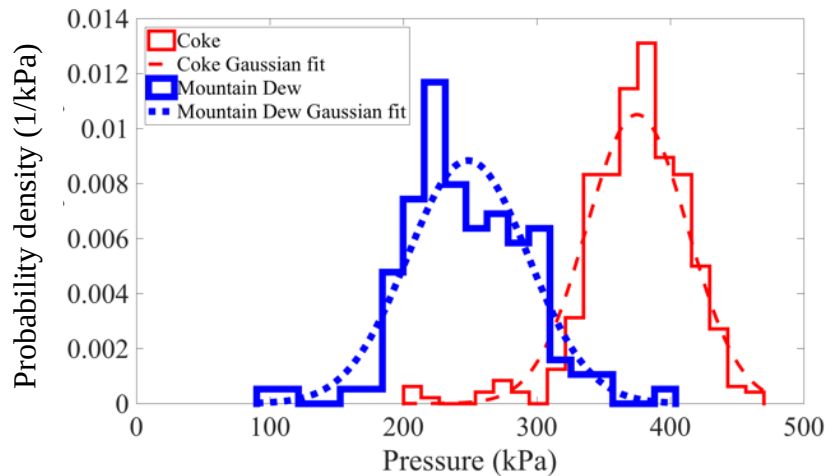


Figure 3: Probability density functions from historical 2.671 data for the internal pressure of Coke ($N = 356$) and Mountain Dew ($N = 120$). The mean pressure for Coke 375.1 ± 4.0 kPa and the mean pressure for Mountain Dew is 248.4 ± 8.2 kPa.

Examining this figure, it seems obvious that Coke and Mountain Dew have different average pressures. Using data from the distributions in Figure 3, `ttest2(coke, mountain_dew)` returns a 2-tailed p-value of $p = 9.56 \times 10^{-67}$. Since $p < 0.05$, you reject the null hypothesis because it is unlikely that Coke and Mountain Dew have the same mean pressure. However, because p is not zero, it is still *possible* (but extremely improbable) that Coke and Mountain Dew do in fact have the same mean pressure.

Consider a less obvious case. Suppose you obtain a p-value of $p = 0.032$. You reject the null hypothesis because $p < 0.05$, but there is still a finite probability that the null hypothesis is actually true. A p-value of 0.05 means there is a 5% chance of getting your observed results with the null hypothesis being true. It **does not mean** that there is a 5% chance that the null hypothesis is true.

Stating that something is statistically “significant” often leads to misunderstandings. As with “probability” and “confidence”, the common use of the word “significance” as a synonym for “importance” can be misleading. **Low p-values are called statistically significant.** Do not confuse “significant” with “important” or “useful”. In statistics, significant means “unlikely to be the result of random chance”, or “likely to be replicated if the procedure is repeated”. Moreover, you may observe a robust finding that is statistically significant, but of limited utility or practical importance. At the conclusion of your statistical analysis, you must exercise engineering judgment to decide what the results mean in context.

2.4 Hypothesized Mean Difference

The Hypothesized Mean Difference can be used to strengthen your conclusions about the difference between two samples. As an example, consider the Coke and Mountain Dew populations shown in Fig. 3. The difference in the mean values is 126.7 ± 8.2 kPa, where we have used the propagation of uncertainty method to determine the 95% confidence interval on the difference in pressure. Given the extremely low p-value, you can state that the internal pressure of

Coca Cola is greater than that of Mountain Dew with an extremely high value of confidence (p-value $\sim 10^{-60}$), but it would be more useful to know **how different** they are. Finding the hypothesized mean difference allows you to make the much stronger statement that “The average pressure of Coca Cola is 119.1 kPa greater than that for Mountain Dew with 95% confidence” rather than the true but not as meaningful statement that “The average pressure of Coca Cola is greater than that of Mountain Dew with $> 99.999999999999\%$ confidence”.

The Hypothesized Mean Difference (HMD) is defined as the **upper limit of the difference in mean of the two populations with 95% confidence**. It is determined by combining the two starting distributions into a third distribution, which is the difference between the first two. The hypothesized mean difference is the value such that the cumulative probability up to that value corresponds with your chosen confidence level. This approach is shown graphically in Fig. 4.

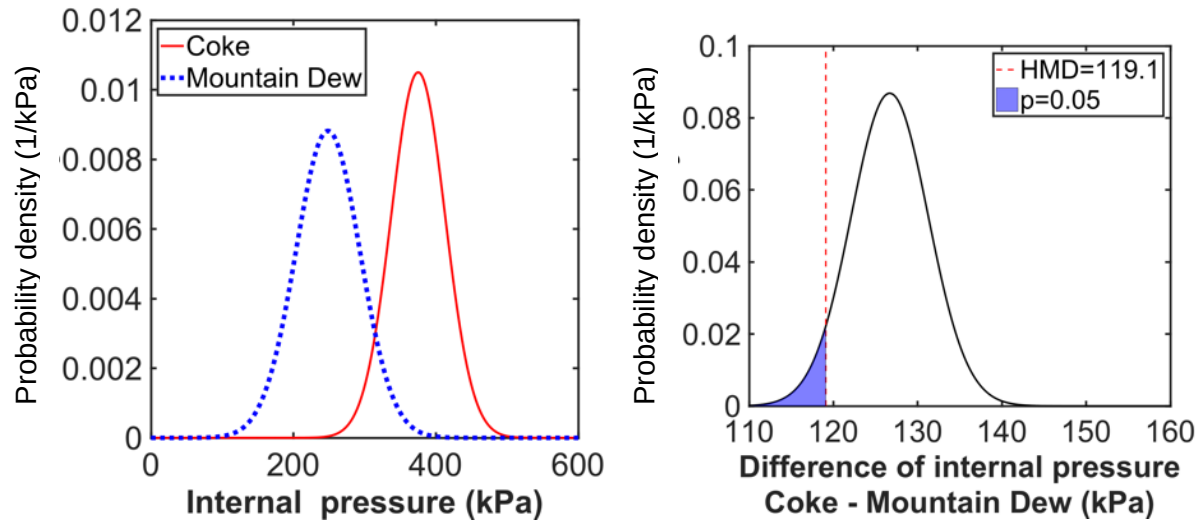


Figure 4: Conversion of the two populations (left graph) into the difference distribution (right graph) allows graphical identification of the HMD as the upper boundary of the region containing 5% of the total area (for 95% confidence, which corresponds to p-value = 0.05).

The HMD shown in the right graph of Fig. 4 of 119.1 kPa is the upper limit of the value we can state with 95% confidence. If you shift the red dashed line to the left (lower pressure difference) you will enclose less than 5% of the area, and thus will have greater than 95% confidence of a 118 kPa difference. Shifting to the right will enclose more than 5% of the area, so that there will be less than 95% confidence of a 120 kPa difference.

There is 95% confidence that that the difference between the two populations is 119.1 kPa. This is a very powerful statement when you are trying to compare the results of two data sets with different conditions!

The formula for the hypothesized mean difference, Δ_{HMD} , is:

$$\Delta_{\text{HMD}} = (\mu_1 - \mu_2) - (t_{\text{conf}, \nu_{12}}) S_{12}, \quad (2)$$

where μ_1 is the mean of the distribution with the larger mean
 μ_2 is the mean of the distribution with the smaller mean

$t_{\text{conf}, \nu_{12}}$ is the 2-tailed t-factor (see Eq. (1) in Sect. 2.1) with ν_{12} degrees of freedom (Eq. (4), based on the Welch–Satterthwaite equation) [You may substitute a 1-tailed t-factor iff you have justification for 1-tail test.]

$$s_{12} = \sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}} \quad (3)$$

$$\nu_{12} = \frac{s_{12}^4}{\frac{s_1^4}{N_1^2 \nu_1} + \frac{s_2^4}{N_2^2 \nu_2}} \quad (4)$$

s_1 , N_1 , and ν_1 are the standard deviation, number of points and degrees of freedom, respectively, (i.e. $\nu_1 = N_1 - 1$) of distribution 1 (the distribution with the larger mean)

s_2 , N_2 , and ν_2 are the standard deviation, number of points and degrees of freedom, respectively, (i.e. $\nu_2 = N_2 - 1$) of distribution 2 (the distribution with the smaller mean)

Equations (2)-(4) may look intimidating. To avoid implementing these complicated formulas, for the Soda Can lab you will use a guess and check method in Excel, or you can use the MATLAB code *HMD_example.m* from the Wiki.

If you are interested in a more in-depth graphical explanation of how the HMD approach allows you to determine an upper limit on the difference between the mean values of two populations, please refer to Appendix A, after the References.

3. ANOVA

The Analysis of Variance (ANOVA) is a statistical test that generalizes and extends the Student's t-test. Where the 2-sample Student's t-test measures whether two samples have a statistically significant difference in their means, ANOVA provides a statistical test of whether two (more more) sample means are equal. It does not quantify the difference in means, but can let you establish that there is a statistically significant difference among groups. It should not be used with ordered data, for example, when the independent variable is elapsed time or distance from the origin. In those cases, traditional regression should be used. Please refer to the code *anova_example.m* on the MATLAB – Uncertainty Calculations page on the 2.671 Wiki.

4. Using Uncertainty Analysis in your Reports

Hopefully the Uncertainty Analysis documents have given you a good introduction to uncertainty analysis so that you will not be tempted to say “the agreement was pretty good” in any of your 2.671 reports. Instead you should say “the result was 3.36 ± 0.12 (units) with 95% confidence, as compared to the expected value of 3.40 (units). There is no significant discrepancy between the measured and expected values.” Or, if the expected value was 3.50 (units), “the expected value is outside the 95% confidence interval of the measurement, and therefore there is a significant discrepancy between the measured and expected values. We hypothesize that this discrepancy is due to but further experiments would be required to test this hypothesis.”

Many of your Go Forth projects will entail comparing measurements for populations under different conditions; for example, comparing the reaction time for swimmers starting from a block

with and without a wedge. If there is a statistically significant difference between measured reaction times, the hypothesized mean difference approach can be used to quantify the difference in start times **either as a ratio or an absolute difference**, depending on which is most relevant to your objective question. If the measured reaction time with a wedge is 0.67 ± 0.02 s, and the reaction time without a wedge is 0.70 ± 0.02 s, and the p-value is less than 0.05 (which can happen even with error bar overlap), you should use the approach outlined above to find the HMD. Let's suppose you find $\text{HMD} = 0.012$ s. You can either report that the decrease in reaction time with a wedge is 0.012 s (with 95% confidence), or that adding a wedge decreases reaction time by 1.7% ($0.012/0.70$).

References

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- [6] Welch's t-test, https://en.wikipedia.org/wiki/Welch%27s_t-test, accessed 2/6/18. (*We are searching for a better reference than Wikipedia, as we realize this is not an acceptable technical reference, 2/6/18*)

APPENDIX A: Graphical description of Hypothesized Mean Difference

Only read this section if you are unsure about the meaning of the Hypothesized Mean Difference as explained in Sect. 2.4, pp. 8 – 9.

Imagine that you shifted the Mountain Dew distribution shown in Fig. 3 to the right by 40 kPa. Now the distributions overlap more, and the peaks are less distinct. If you run the 2-sample t-test comparing Coca Cola and the shifted Mountain Dew distribution, you would get a 1-tail p-value of 8.21×10^{-45} . That is such a small probability, we can use a 1-tail HMD in further analysis.

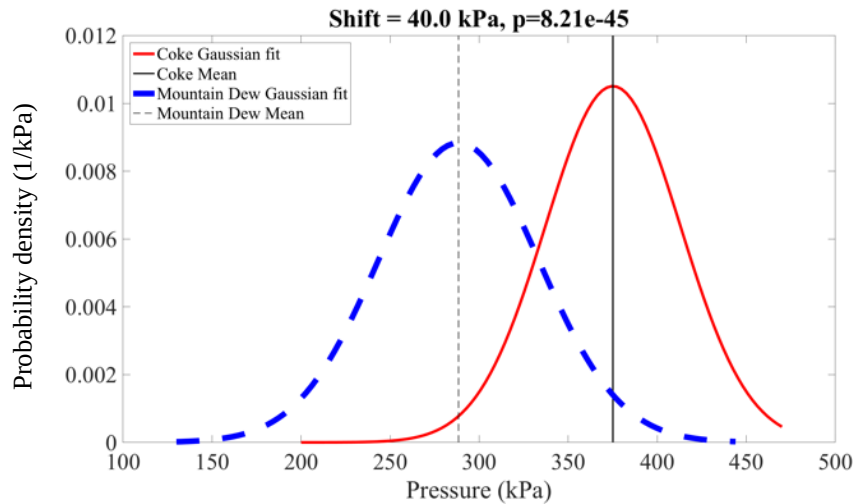


Figure 5: Comparison of Coke and Mountain Dew after shifting the Mountain Dew population to the right (higher pressure) by 40 kPa. The resultant one-tailed p-value is 8.21×10^{-45} .

This p-value is smaller than before, which indicates there is a greater (although still infinitesimal) chance that the observed distributions were sampled from one common parent distribution just by chance. Likewise, shifting by 80 kPa gives a p-value of 8.69×10^{-20} .

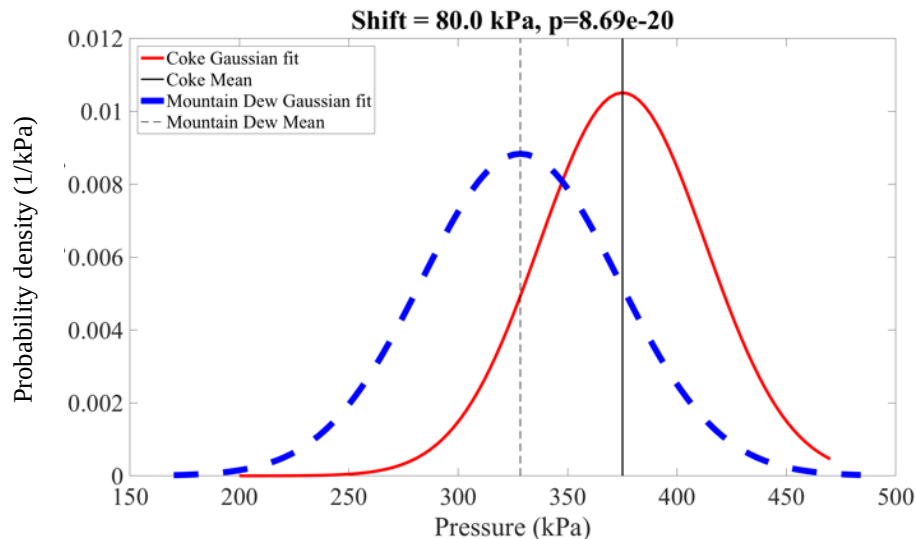


Figure 6: Comparison of Coke and Mountain Dew after shifting the Mountain Dew population to the right (higher pressure) by 80 kPa. The resultant one-tailed p-value is 8.69×10^{-20} . There is still a negligible chance that the two population means are the same.

In other words, we are confident that not only are the internal pressures of Coca Cola and Mountain Dew different, but also that they differ by at least 80 kPa. It would be very improbable to start over and repeat our sampling procedure to find results as extreme as this purely by chance.

How does this test work? This test combines the two starting distributions into a new distribution. Assume μ_1 is the larger mean. The new distribution has mean of $(\mu_1 - (\mu_2 + \text{HMD}))$, where HMD is the shift, or hypothesized mean difference. The variance of the combined distribution is approximated based on the variance and degrees of freedom of the starting distributions. [5] Then, with this new distribution and a specified confidence level, find the maximum value of HMD that would still exclude zero from the confidence interval. We are answering the question “How much can we nudge the distributions together and still state they are different with the confidence level we want?”. The farther apart the distributions are to begin with, the more confidence we have that they are really different (low p-value) and the more we can nudge them together. The closer we nudge them together, the greater a difference we can assert between the starting distributions, but with lower confidence.

If you shift the Mountain Dew distribution up by 119.1 kPa, you achieve a p-value of 0.05. **This means you can state with 95% confidence that the internal pressure of Coke is 119.1 kPa greater than the internal pressure of Mountain Dew.**

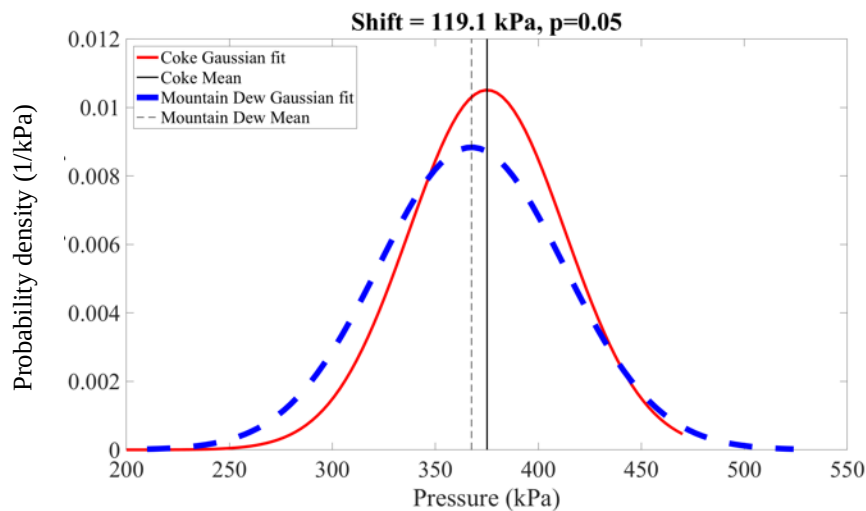


Figure 7: Shifting the Mountain Dew population up by 119.1 kPa results in a p-value of 0.05, which means that with 95% confidence, the internal pressure of Coke is 119.1 kPa greater than the internal pressure of Mountain Dew.

If you shift the Mountain Dew distribution up by 121 kPa, you would achieve a p-value of 0.11, which means that you have **less than 90% confidence** that the mean pressure of Coke is 121 kPa greater than the mean pressure of Mountain Dew.

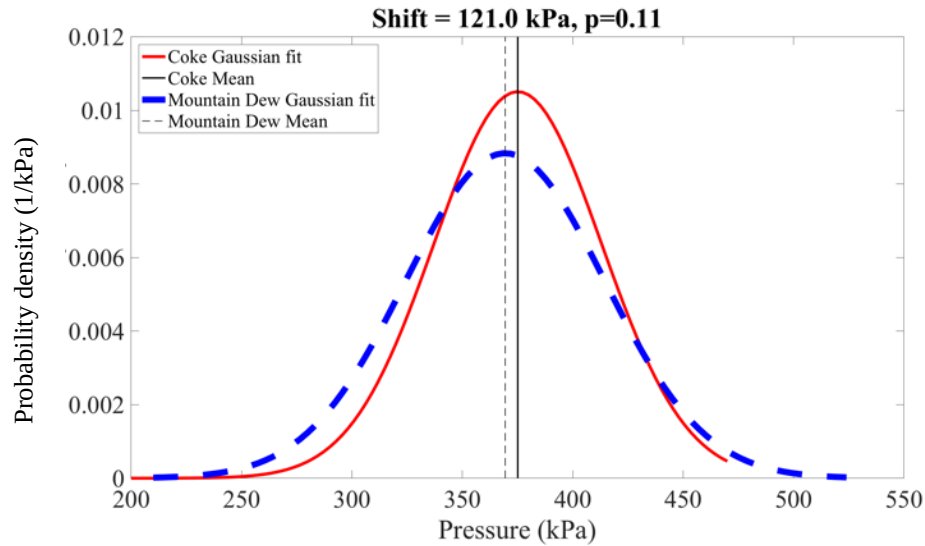


Figure 8: Increasing the shift to 121.0 kPa results in a p-value of 0.11, which means that you have less than 90% confidence that the internal pressure of Coke is 121.0 kPa greater than the internal pressure of Mountain Dew.

Therefore, as formulated here, where we shifted the distribution with the smaller mean towards the distribution with the greater mean, the hypothesized mean difference that produces a p-value of 0.05 is the **upper limit** of the value we can state with 95% confidence. We have greater than 95% confidence of a 118 kPa difference, 95% confidence of a 119 kPa confidence, and less than 95% confidence of a 120 kPa difference. This is a very powerful statement when you are trying to compare the results of two data sets with different conditions!